

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary End Semester Examination – Summer Semester 2023**

**Course: B. Tech.**

**Branch: Computer /CSE**

**Semester: V**

**Subject Code & Name: (BTCOC501) Database System**

**Max Marks: 60**

**Date:07/08/2023**

**Duration: 3 Hr.**

**z Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO)    Marks

**Q. 1 Solve any two of the following.**

- |                                                                                              |            |            |
|----------------------------------------------------------------------------------------------|------------|------------|
| <b>A) Illustrate the database characteristics. How they are different from File systems.</b> | <b>L 2</b> | <b>6 M</b> |
| <b>B) Illustrate about integrity and key constraints with suitable examples?</b>             | <b>L3</b>  | <b>6 M</b> |
| <b>C) Explain the importance of Null values in Relational Model.</b>                         | <b>L2</b>  | <b>6 M</b> |

**Q.2 Solve all two of the following.**

- |                                                                                                                |           |            |
|----------------------------------------------------------------------------------------------------------------|-----------|------------|
| <b>A) Explain in detail, the form of a basic SQL query with a suitable example.</b>                            | <b>L2</b> | <b>6 M</b> |
| <b>B) List out various SET comparison operator in SQL and also write about its use in writing SQL queries.</b> | <b>L3</b> | <b>6 M</b> |

**Q. 3 Solve any two of the following.**

- |                                                              |           |            |
|--------------------------------------------------------------|-----------|------------|
| <b>A) Write SQL Queries for the following set of tables:</b> | <b>L3</b> | <b>6 M</b> |
|--------------------------------------------------------------|-----------|------------|

account(account\_number, branch\_name, balance)  
branch (branch\_name, branch\_city, assets)  
customer (customer\_name customer\_street, customer\_city)  
loan (loan\_number, branch\_name, amount)  
depositor((customer\_name, account\_number)  
borrower(customer\_name, loan\_number)

- a. For all customers who have a loan from the bank, find their names and loan numbers with the attribute loan\_number replaced by loan\_id.
- b. Find the names of all customers whose street address includes the substring 'Main'.
- c. List loan data, ordered by decreasing amounts, then increasing loan numbers.

**B) Write SQL Queries for the following set of tables: L3 6 M**

EMPLOYEE (FNAME, MINIT, LNAME, SSN, BDATE, ADDRESS, SEX, SALARY, #SUPERSSN, #DNO),  
 DEPARTMENT (DNAME, DNUMBER, #MGRSSN, MGRSTARTDATE),  
 DEPT\_LOCATIONS (#DNUMBER, DLOCATION),  
 PROJECT (PNAME, PNUMBER, PLOCATION, #DNUM),  
 WORKS\_ON (#ESSN, #PNO, HOURS),  
 DEPENDENT (#ESSN, DEPENDENT\_NAME, SEX, BDATE, RELATIONSHIP)

- a. Retrieve the name and address of all employees who work for the 'Research' department
  - b. Retrieve the birthdate and address of the employee(s) whose name is 'Raj V Naik'
  - c. Retrieve all the attributes of an EMPLOYEE and the attributes of the DEPARTMENT he or she works in for every employee of the 'Research' department
- C) By considering an example describe various data update operations in SQL. L2 6 M**

**Q.4 Solve all two of the following.**

- A) State BCNF. How does it differ from 3NF? L2 6 M**
- B) Explain insertion, deletion and modification anomalies with suitable examples. L2 6 M**

**Q. 5 Solve any two of the following.**

- A) Explain in detail about internal hashing Techniques. L2 6 M**
- B) Discuss in detail about cluster and Multilevel indexes. L2 6 M**
- C) Explain Concurrency control with locking methods. L2 6 M**

**\*\*\* End \*\*\***